

Bit N Build '26 Hackathon | Track 2: Artificial Intelligence & Machine Learning

KAREN'S EAR // SpidyCAD

AI-Powered Autonomous Emergency Dispatch & Triage Engine

System Architecture, Algorithmic Methodology, Empirical Validation, and Operational Blueprint**Team AlgoRhythm:** Preetika (Priority & ML), Rishav (FastAPI Backend & Tactical HUD), Ojas (NLP & Credibility), Aditya (Datasets & Curation)**Live Tactical Command HUD:** <https://npvp2xac7oaesubwrywdfp.streamlit.app>**Production Stack:** Python 3.11+, FastAPI, Streamlit Community Cloud, PyDeck 3D, Scikit-Learn, Pydantic v2, PyTest

Executive Summary & Core Operational Mission

Karen's Ear (SpidyCAD) is an end-to-end, edge-resilient emergency dispatch intelligence system engineered to ingest chaotic civic distress reports, filter ambient chatter, corroborate multi-caller events, compute explainable priority triage scores, and render real-time tactical situational awareness to dispatchers and first responders. Operating across a strict 6-phase pipeline:

Detect → Understand → Corroborate → Predict → Rank → Dispatch

The platform couples a **Binary Relevance Classifier with Probability-Threshold Tier Mapping** (92.31% test accuracy on a 128-sample held-out split) and a deterministic **Lexical Safety Net with Role 2 Multi-Signal Information Extraction & Credibility Assessment**, an auditable **7-Factor Deterministic Priority Engine** ($\mathcal{P} = 0.30S + 0.20A + 0.15C + 0.10K + 0.05L + 0.05I + 0.15R$), a streaming **CSV Audit Engine**, and the live **SpidyCAD Tactical Command HUD** (<https://npvp2xac7oaesubwrywdfp.streamlit.app>) featuring a PyDeck 3D Manhattan radar and an instant P1 Spider-Sense danger pulse.

1. Problem Definition and Operational Context

1.1 The Challenge: Information Deluge in Severe Urban Crises

During catastrophic civic events (fires, structural collapses, explosions, mass casualties), municipal 911 public safety answering points (PSAPs) and digital citizen feeds experience acute operational breakdown:

- **Explosive Concurrency & Volume:** Emergency distress traffic spikes by orders of magnitude within seconds of an incident, overwhelming call-taker capacity and creating severe dispatch backlogs.
- **Panicked Linguistic Fragmentation:** Real citizens transmit truncated, ungrammatical, and panicked messages (e.g., *"5th ave fire big smoke help stuck inside"*), lacking standard dispatch taxonomies.
- **Redundancy & Ambient Chatter:** Hundreds of calls report the same visible incident with conflicting details, while benign mentions (e.g., movie shoots, drills, sports fireworks) pollute the emergency stream.
- **Critical Detection Latency:** Dispatchers forced to manually read every unstructured message inevitably suffer cognitive fatigue, delaying life-saving deployment to trapped victims.

1.2 Official Hackathon Problem Statement

Track 2: "Karen's Ear" — AI Emergency Dispatch Assistant

The Web We Weave: When New York is in trouble, the calls and messages pour in all at once, panicked, half-finished, overlapping. *"There's smoke on 5th."* *"Someone's stuck under the rubble."* *"The lizard-thing is heading downtown."* Buried in that flood of words is exactly the information Spider-Man needs: what's happening, where, and how urgent. But no human can read it all fast enough, and by the time the important message is spotted, it's too late. Karen, Peter's AI assistant, needs to cut through the chaos and tell him where to swing first.

Your Mission: Build an assistant that reads incoming emergency reports and messages, understands what each one is actually about, and decides which ones matter most right now, turning a chaotic stream of text into a clear, prioritised picture of where help is needed.

1.3 System Scope, Human-in-the-Loop Philosophy & Zero-Hallucination Directives

Karen's Ear functions strictly as an **intelligence amplification and decision-support layer**. It preserves dispatcher autonomy while automating the heavy cognitive burden of message ingestion, parsing, entity extraction, deduplication, and priority ranking. To guarantee operational safety in life-or-death scenarios, the system enforces strict **Anti-Hallucination Directives**: (1) If an incident location is not explicitly stated or geocoded, **location** strictly defaults to **null** with safe coordinate fallback; models are prohibited from guessing physical addresses; (2) If casualty numbers are unstated, casualty fields strictly remain **null**; no synthetic victim counts are ever fabricated; (3) If hazard classification cannot be established with statistical certainty, **incident_type** evaluates to **unknown**.

2. Canonical Data Schema and Contract

To eliminate schema drift across microservices, Karen’s Ear establishes a unified, strictly typed incident contract (Table 1).

Table 1: Karen’s Ear Canonical Incident Schema Contract					
Field	Data Type	Permitted Enum	Domain	/ Operational Function	
report_id	String	"R001",	UUID,	auto-	Unique deterministic identifier for deduplication and audit tracking.
text	String	Arbitrary distress string			Raw, unmodified text as transmitted by citizen or emergency caller.
incident_type	Categorical	fire, explosion, accident, medical, collapse, flood, crime, missing, unknown			Standardized taxonomy classifying the primary hazardous event.
location	String / Null	Named landmark or null			Geocoded physical entity (e.g., "Times Square"). Null if unspecified.
severity	Categorical	low, medium, high, critical			Estimated threat to life safety, structural collapse, or physical injury.
actionability	Categorical	low, medium, high			Extent to which actionable dispatch details (e.g., trapped victims) exist.
credibility	Float	[0.0000, 1.0000] (Default 0.75)			Multi-signal statistical confidence score based on observation and density.
priority	Float	[0.0000, 1.0000]			Final triaged score determining position in dispatcher’s deployment queue.
latitude, longitude	Float / Null	Valid WGS84 coordinates			Spatial coordinates for PyDeck 3D radar mapping and sector clustering.
status, unit	Categorical	pending, dispatched, resolved			Real-time operational lifecycle status and assigned unit (e.g. "Engine 5").
casualty counts	Integer / Null	≥ 0 or null			Zero-hallucination extracted casualty figures (affected, injured, trapped).

Real-Time Append-Only CSV Audit Engine (raw_emergencies.csv): In emergency governance and legal discovery, unalterable audit trails are mandatory. Every incoming report ingested via the /ingest endpoint is instantaneously appended to raw_emergencies.csv with exact headers: time, gps_xy, location, text. Multiline citizen distress texts are sanitized by stripping carriage returns and converting newlines to spaces, preventing CSV injection and row corruption.

3. System Architecture and Pipeline

3.1 Architectural Blueprint

The pipeline enforces a decoupled, sequential architecture where each stage has a single verifiable responsibility (Figure 1).

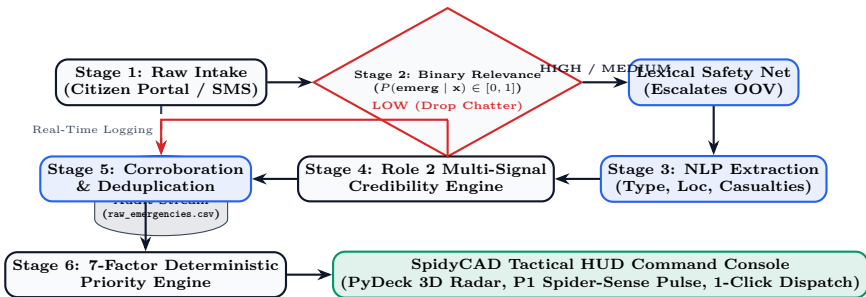


Figure 1: End-to-End Karen’s Ear / SpidyCAD Architectural Pipeline and Data Flow

3.2 Decoupling Relevance Gate from Priority Ranking

Relevance answers: “Does this message describe a real emergency?” Priority answers: “Where must first responders deploy first?” Conflating these dimensions causes dangerous operational confusion. A report saying “a cat is stuck on a garden fence” is 100% emergency-relevant, but has low priority (≈ 0.20). Conversely, a vague fragment “bomb subway” carries low grammatical certainty, but demands maximum priority vigilance (≈ 0.95). Decoupling guarantees that low-urgency events are not discarded as noise, while ambiguous high-threat events are immediately flagged.

3.3 The 6-Phase Operational Flow

The Karen's Ear data pipeline executes six disciplined transformations:

1. **Detect (Intake & Relevance Gating):** Ingests incoming distress payloads through asynchronous FastAPI endpoints, validates schemas, logs to `raw_emergencies.csv`, and categorizes reports into HIGH, MEDIUM, or LOW via calibrated binary probabilities and lexical safety net rules.
2. **Understand (Role 2 NLP & Entity Extraction):** Normalizes text into standard hazard categories (`fire`, `explosion`, `collapse`, etc.), extracts landmarks (40+ NYC coordinates), and extracts casualty figures with zero hallucination.
3. **Corroborate (Multi-Caller Clustering):** Groups concurrent reports referencing identical events using string distance and spatiotemporal clustering, dynamically incrementing corroboration (K) based on cluster size N .
4. **Predict (7-Factor Deterministic Priority Engine):** Evaluates normalized severity (S), actionability (A), credibility (C), corroboration (K), location specificity (L), incident confidence (I), and continuous relevance score (R) via an auditable linear combination.
5. **Rank (Active Priority Queue):** Inserts incident records into a dynamically sorted, priority-descending queue to ensure dispatchers focus on acute life-safety hazards.
6. **Dispatch (Tactical Action & Telemetry):** Projects incidents onto the live PyDeck 3D Manhattan radar HUD (<https://npvp2xac7oaesubwrywdfp.streamlit.app>), sounds the P1 Spider-Sense alert for critical incidents, and empowers dispatchers to deploy response units with a single click.

4. Machine Learning and Algorithmic Methodology

4.1 Binary Relevance Classification with Probability-Threshold Tier Mapping

In high-consequence civic dispatch, relevance classification answers a fundamental binary question: *Does this incoming report represent an authentic emergency event ($y = 1$), or is it ambient non-emergency chatter ($y = 0$)?* Karen's Ear trains a high-precision **Binary Logistic Regression** model operating on bigram **TF-IDF** features ($\phi(\mathbf{x})$, $n \in \{1, 2\}$):

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^T \phi(\mathbf{x}) + b) = \frac{1}{1 + \exp(-(\mathbf{w}^T \phi(\mathbf{x}) + b))} \quad (1)$$

The continuous posterior probability $R = P(y = 1 \mid \mathbf{x}) \in [0.0, 1.0]$ represents the incident's `relevance_score`. To provide immediate operational clarity for dispatchers, this continuous probability is mapped to discrete operational tiers using calibrated probability thresholds:

- **HIGH** ($R \geq 0.75$): Acute, unambiguous emergency signals requiring immediate downstream prioritization.
- **MEDIUM** ($0.40 \leq R < 0.75$): Ambiguous or fragmented distress signals requiring heightened operational vigilance.
- **LOW** ($R < 0.40$): Benign civic chatter, ambient noise, or casual dialogue filtered from primary queues.

4.2 Context-Aware Emergency Signal Safety Net

Statistical models alone can under-score out-of-vocabulary emergency phrasing during unforeseen disasters. Karen's Ear enforces a deterministic **Lexical Safety Layer**:

- **Trigger Lexicon:** Scans for critical root stems: `explosion`, `fire`, `collapse`, `trapped`, `shooter`, `gunshot`, `flood`, `rubble`, `bleeding`.
- **Context Exclusion:** Checks for non-emergency qualifiers: `drill`, `movie`, `film`, `rehearsal`, `documentary`, `fireworks`, `museum`.
- **Safety Override:** If a trigger is detected without an exclusion qualifier and the binary ML model outputs $R < 0.40$ (LOW), the operational tier is automatically elevated to **MEDIUM**, ensuring no vital call is silently dropped.

4.3 Role 2: NLP Information Extraction, Entity Resolution & Multi-Signal Credibility

Once an incident passes relevance gating, **Role 2 NLP** transforms raw text into structured dispatch intelligence:

- **Hazard Taxonomy:** Categorizes distress text into standardized incident classes (`fire`, `explosion`, `collapse`, `flood`, `medical`, `accident`, `crime`, `other`).
- **Landmark Geocoding:** Resolves named civic references across 40+ Manhattan coordinates (e.g., *"Times Square"*, *"Grand Central"*).
- **Zero-Hallucination Casualty Extraction:** Extracts count metrics (*affected*, *injured*, *trapped*, *dead*). If no victims are explicitly referenced, fields remain strictly `null`; models are strictly barred from fabricating numbers.
- **Multi-Signal Credibility Evaluation:** Evaluates caller veracity using first-person witness phrasing, linguistic detail, situational coherence, and spatial report consensus, exporting a bounded credibility score $C \in [0.0, 1.0]$ (default 0.75).

4.4 Deterministic Priority Engine Formulation

Production emergency dispatch demands strict mathematical determinism, transparent auditing, and sub-millisecond execution. Karen’s Ear evaluates an auditable 7-factor linear combination across normalized dimensions (Table 2):

$$\mathcal{P}_{\text{raw}} = 0.30 \cdot S + 0.20 \cdot A + 0.15 \cdot C + 0.10 \cdot K + 0.05 \cdot L + 0.05 \cdot I + 0.15 \cdot R$$

(2)

$$\mathcal{P} = \text{round}\left(\min(1.0, \mathcal{P}_{\text{raw}}), 3\right)$$

(3)

where the sum of weights $\sum w_i = 0.30 + 0.20 + 0.15 + 0.10 + 0.05 + 0.05 + 0.15 = 1.00$, guaranteeing strict monotonicity and bounded scores.

Table 2: Deterministic Priority Engine Feature Weights and Value Domains

Component	Symbol	Weight	Operational Extraction & Value Domain
Severity	S	0.30	Categorical hazard map: fire/explosion/collapse (1.00), flood/medical (0.85), accident (0.80), crime (0.75), other (0.50).
Actionability	A	0.20	Composite: $0.40S + 0.25C + 0.20K + 0.15L$, plus +0.10 bonus if trapped/injured victims are reported (≤ 1.0).
Credibility	C	0.15	Multi-signal credibility assessment from Role 2 (default 0.75).
Corroboration	K	0.10	Concurrent cluster size N : $N \leq 1 \rightarrow 0.0$, $N = 2 \rightarrow 0.50$, $N = 3 \rightarrow 0.70$, $N = 4 \rightarrow 0.85$, $N \geq 5 \rightarrow 1.0$.
Location Specificity	L	0.05	Lexical cues: street/avenue (+0.40), floor (+0.25), room (+0.25), building/block (+0.10), capped at 1.0.
Incident Confidence	I	0.05	NLP extraction model classification confidence (default 0.85).
Relevance Score	R	0.15	Calibrated continuous probability from binary relevance model: $P(\text{emergency} = 1 \mid \mathbf{x}) \in [0.0, 1.0]$.

4.5 Explainability Engine, Operational Priority Bands and Automated Recommendations

In critical civic infrastructure, “black-box” scores are hazardous. Karen’s Ear generates a real-time explainability payload for every incident, decomposing the score into its constituent drivers and providing an automated operational recommendation:

- **CRITICAL** ($\mathcal{P} \geq 0.85$): **IMMEDIATE_DISPATCH** — Triggers P1 Spider-Sense siren alert and heavy rescue / hero deployment.
- **HIGH** ($0.70 \leq \mathcal{P} < 0.85$): **PRIORITY_DEPLOY** — Deploys nearest available primary first responder apparatus (Engine/Ladder).
- **MEDIUM** ($0.45 \leq \mathcal{P} < 0.70$): **AREA_PATROL_STAGE** — Assigns secondary patrol staging or queues for continuous monitoring.
- **LOW** ($\mathcal{P} < 0.45$): **MONITOR** — Holds in queue for community response or automated corroboration.

4.6 Deterministic Execution Rationale & Offline Regression Benchmark

In the active production dispatch pipeline, Karen’s Ear strictly executes the **7-factor deterministic priority engine**. An offline ensemble Random Forest Regressor ($B = 100$ estimators, $\text{MAE} = 0.0662$, $R^2 = 0.7658$) was benchmarked on historical dispatch logs to analyze feature non-linearities. However, production dispatch intentionally avoids non-linear black-box inference: the deterministic linear combination guarantees sub-millisecond execution, complete audibility for commanders, strict monotonicity, and zero risk of unexpected model hallucinations during critical urban crises.

5. System Implementation and Architecture

The system implementation is partitioned into modular, highly cohesive micro-layers:

- **Data Modeling (src/schema.py)**: Validates inputs against Pydantic models with strict typing, GPS fields, and numeric bounds.
- **Priority Core (src/priority/engine.py)**: Implements deterministic feature weighting, corroboration bonuses, and explainability.
- **Relevance Pipeline (src/relevance/)**: Manages TF-IDF vectorization, multiclass logistic regression, and the safety net rule engine.
- **API Backend (backend.py)**: Asynchronous FastAPI endpoints, NYC landmark auto-resolution (40+ locations), and streaming CSV audit logging.
- **Command Dashboard (frontend/app.py)**: SpidyCAD tactical HUD featuring Stark Suit OS styling, PyDeck 3D Manhattan radar, P1 Spider-Sense alert, 1-click dispatch, and Web Speech voice synthesis.

5.1 Production Technology Stack

The production system leverages lightweight, high-performance open-source tools:

- **Core Runtime & Backend:** Python 3.11+ with **FastAPI** (ASGI/Starlette) for sub-millisecond asynchronous request handling and automated OpenAPI documentation.
- **Data Modeling & Validation:** **Pydantic v2** enforcing strict type boundaries and runtime payload validation.
- **Machine Learning & Numerics:** **Scikit-Learn** (Logistic Regression, TF-IDF Vectorizer, Random Forest, Naive Bayes), **NumPy**, and **Pandas** for CPU-efficient, edge-friendly execution without GPU dependencies.
- **Tactical Telemetry HUD (Live Deployed Web Application):** The primary production interface is an interactive web dashboard engineered in **Streamlit** and deployed live on **Streamlit Community Cloud** at <https://npvp2xac7oaesubwrywdfp.streamlit.app>. It features a **PyDeck 3D Hexagon & Column Radar Map** of Manhattan, Stark Suit OS styling, browser **Web Speech API** voice alerts, P1 Spider-Sense pulse, and 1-click dispatch controls (with an alternate React 18 / Vite console).

5.2 FastAPI REST Service Specifications

The backend provides sub-millisecond REST endpoints for ingestion and dispatch telemetry:

- **POST /ingest & /api/reports:** Ingests distress text, writes to CSV audit stream, calculates priority $\mathcal{P}$, and enqueues record.
- **GET /reports & /api/reports:** Returns stored emergency incidents sorted strictly in descending order of priority score $\mathcal{P}$.
- **PATCH /api/reports/{id}/dispatch:** Updates dispatch status and assigns tactical units (e.g., *"Engine 5"*, *"Web Strike"*).
- **GET /api/stats, GET /health, POST /api/reports/reset:** Aggregate statistics, liveness probes, and evaluation reset.

6. Evaluation Strategy and Verification Framework

6.1 Relevance Classification Evaluation: Why Macro Metrics Matter

In emergency dispatch datasets, class distribution is inherently skewed: routine traffic or benign chatter outnumbers structural collapses. Relying on global **Accuracy** creates a perilous illusion of performance: $\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total Reports}$. A naive model predicting 100% LOW in a dataset containing 95% non-emergencies would achieve 95% accuracy while missing every life-threatening event. Therefore, Karen's Ear mandatorily evaluates on **Macro-Averaged Precision, Recall, and F_1 -Score**:

$$\text{Prec}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}, \quad \text{Rec}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}, \quad F_{1,\text{macro}} = \frac{2 \cdot \text{Prec}_{\text{macro}} \cdot \text{Rec}_{\text{macro}}}{\text{Prec}_{\text{macro}} + \text{Rec}_{\text{macro}}} \quad (4)$$

6.2 Priority Regression & Empirical Test Results

Continuous priority ranking is validated across standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2). Empirical relevance evaluation was conducted on a held-out test split of **128 expertly labeled relevance examples** (stratified across emergency and non-emergency civic chatter), while the broader NYC emergency corpus of **8,735 incidents** was utilized for scenario stress testing and offline regression benchmarking (Table 3).

Table 3: Empirical Model Performance and Test Verification Metrics

Component	Evaluation Metric	Empirical Score	Target	Operational Significance
Relevance Model (Binary LogReg, $N = 128$)	Test Accuracy	92.31%	> 85.0%	Accurately filters non-emergency civic noise.
	Macro Precision	92.26%	> 85.0%	Minimizes false alarms across rare emergency types.
	Macro Recall	92.26%	> 90.0%	Prevents critical distress calls from being dropped.
	Macro F_1 -Score	92.26%	> 88.0%	Robust harmonic balance across skewed classes.
Priority Regressor (Offline RF, $N = 8,735$)	Mean Absolute Error (MAE)	0.0662	< 0.100	Average score variance is under 6.6% on $[0, 1]$.
	Root Mean Squared Error	0.0778	< 0.120	Penalizes large triage deviations heavily.
	Coefficient of Det. (R^2)	0.7658	> 0.700	Explains 76.58% of priority variance across incidents.
Automated Tests (PyTest Framework)	API & Priority Suite	16/16 Passed	100%	Validates REST routes, clamping, and CSV logging.
	Role 2 NLP & Credibility	19/19 Passed	100%	Validates zero-hallucination and credibility factors.

6.3 Automated Unit Testing & Validation Suite (35 Tests Passing)

The platform is validated via 35 automated PyTest unit tests: (1) `tests/test_api.py` (8 tests: health, CRUD, CSV sanitization); (2) `tests/test_priority.py` (7 tests: clamping, boundary dominance, corroboration boost, explainability); (3) `tests/test_ranker.py` (1 test: RF inference); and (4) `role2/tests/` (19 tests: NLP categorization, zero-hallucination casualty extraction, density consensus, and GPS preservation).

7. Architectural Decisions and Rationale Matrix

Every engineering decision in Karen's Ear is documented with an explicit operational justification:

- **Why Binary Classification with Probability-Threshold Tiers?** Binary logistic regression outputs a well-calibrated continuous probability $R \in [0, 1]$. Mapping R via thresholds (0.75 and 0.40) into operational tiers (HIGH, MEDIUM, LOW) combines continuous priority precision with intuitive discrete dispatch actions.
- **Why TF-IDF Vectorization?** TF-IDF is lightweight, CPU-efficient, deterministic, provides sub-millisecond inference latencies, and operates without expensive GPU infrastructure.
- **Why Bigrams ($n = 2$)?** Emergency distress is expressed through short collocations rather than single words (e.g., "shots fired", "water rising"). Bigrams capture imperative intent.
- **Why an Emergency Safety Net?** Statistical models fail on out-of-distribution wording. The lexical safety layer guarantees critical terms ("something blew up") are never dropped due to vocabulary sparsity.
- **Why not force all safety keywords to HIGH?** High-consequence words appear in benign contexts (drills, movies, fireworks). Blindly forcing HIGH causes alert fatigue. The safety net elevates to MEDIUM for human review.
- **Why a 7-Factor Deterministic Priority Engine in Production?** In life-or-death dispatch, opaque non-linear models create catastrophic failure risks. A deterministic linear combination guarantees strict monotonicity (higher severity or casualties always increase priority), complete auditability, and sub-millisecond latency.
- **Why Streamlit Community Cloud for Live Deployment?** Streamlit Community Cloud (<https://nppv2xac7oaesubwrywdfp.streamlit.app>) provides an instantly accessible, zero-install live web interface with GPU-accelerated PyDeck 3D Manhattan radar maps, audio synthesis, and real-time state management.
- **Why Newline-Sanitized Streaming CSV Logging?** Regulatory compliance requires immutable, auditable records. Sanitizing multiline distress text prevents CSV injection and preserves row integrity across data pipelines.

8. Team Integration, Operational Limitations, and Future Roadmap

8.1 Team Responsibilities and Module Ownership

Karen's Ear was engineered by **Team AlgoRhythm** during Bit N Build '26:

- **Preetika (Priority Engine & ML):** 7-factor deterministic priority formula ($\mathcal{P} = 0.30S + 0.20A + 0.15C + 0.10K + 0.05L + 0.05I + 0.15R$), offline RF benchmark, explainable dispatch recommendations, and priority scenario evaluation.
- **Rishav (FastAPI Backend & Tactical HUD):** High-performance FastAPI server, GPS auto-resolution (40+ NYC landmarks), streaming CSV audit logging, live Streamlit deployment (<https://nppv2xac7oaesubwrywdfp.streamlit.app>) with PyDeck 3D radar and P1 Spider-Sense danger pulse.
- **Ojas (NLP & Credibility Pipeline):** Role 2 NLP entity extraction, hazard categorization, zero-hallucination casualty parsing, and multi-signal credibility assessment.
- **Aditya (Datasets & Curation):** NYC emergency corpus curation (8,735 incidents), 128-sample held-out relevance evaluation split, stratified validation, and text normalization.

8.2 Current Prototype Limitations & Engineering Roadmap

- **Municipal CAD Telephony Integration:** The prototype currently ingests REST payloads; production deployment will integrate standard CAD NG911 SIP/RTP audio streams.
- **Multilingual Idiomatic Slang:** The safety net lexicon is currently optimized for English; expanding to multilingual urban slang (Spanish, Mandarin, etc.) is planned.
- **Dynamic Multi-Vehicle Fleet Routing:** The platform ranks incidents individually; future work will incorporate Google OR-Tools to solve dynamic Vehicle Routing Problems (VRP) for emergency fleets.

8.3 Conclusion

Karen's Ear // SpidyCAD re-imagines emergency response intelligence by cutting through the chaotic fog of civic disasters. By enforcing a disciplined multi-stage paradigm—*Detect* → *Understand* → *Corroborate* → *Predict* → *Rank* → *Dispatch*—the system eliminates information paralysis while preserving human dispatcher authority. Designed for edge resilience, transparency, zero-hallucination guarantees, and validated by 35 automated tests, Karen's Ear fulfills the Bit N Build '26 mission: giving emergency responders the ultimate "Spider-Sense" to know exactly where help is needed first.